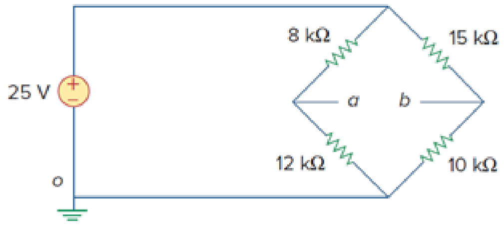
Chapter 2, Problem 70P

Problem

(a) Consider the Wheatstone bridge shown in Fig. 2.130. Calculate *va*, *vb*, and *vab*.

(b) Rework part (a) if the ground is placed at *a* instead of *o*.

Figure 2.130



Step-by-step solution

Step 1 of 6

(a)

Consider the following circuit diagram:

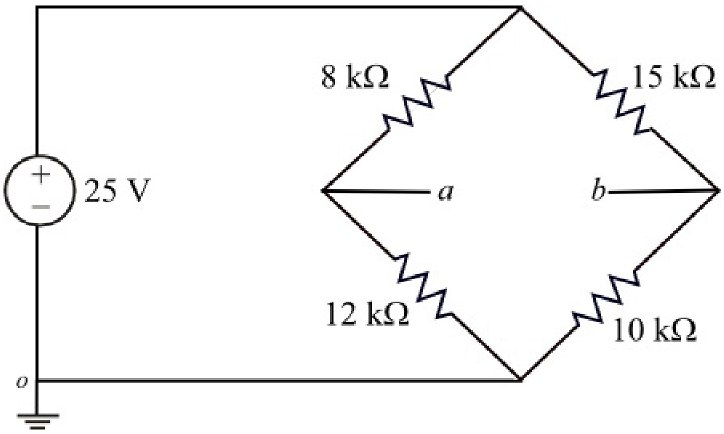


Figure 1

Step 2 of 6

Apply Kirchhoff's current law node *a*.

$$\frac{v_a - 25}{8\times10^3} + \frac{v_a}{12\times10^3} = 0$$
$$\frac{3v_a - 75 + 2v_a}{24\times10^3} = 0$$
$$5v_a - 75 = 0$$
$$v_a = 15\text{ V}$$

Therefore, the voltage at node '*a*', v_a is $\boxed{15\text{ V}}$.

Step 3 of 6

Apply Kirchhoff's current law node *b*.

$$\frac{v_b - 25}{15\times10^3} + \frac{v_b}{10\times10^3} = 0$$
$$\frac{2v_b - 50 + 3v_b}{30\times10^3} = 0$$
$$5v_b - 50 = 0$$
$$v_b = 10\ \text{V}$$

Therefore, the voltage at node '*b*', v_b is $\boxed{10\ \text{V}}$.

Calculate the voltage between node a and b.

$$\begin{aligned} v_{ab} &= v_a - v_b \\ &= 15 - 10 \\ &= 5\ \text{V} \end{aligned}$$

Therefore, the voltage between node a and b, v_{ab} is $\boxed{5\ \text{V}}$.

Step 4 of 6

(b)

Consider the following circuit diagram with interchange ground from *o* to *a*.

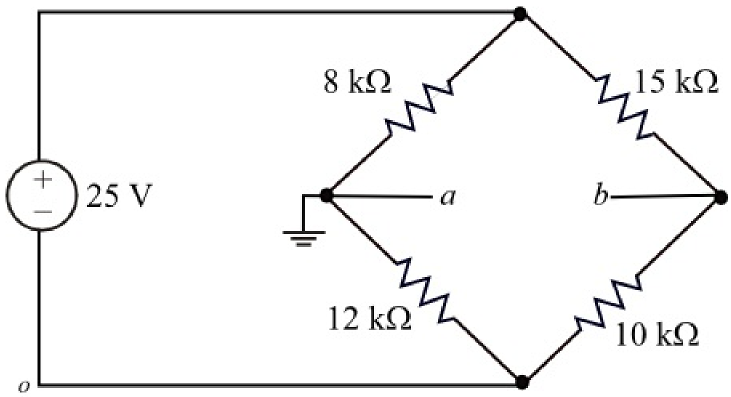


Figure 2

Step 5 of 6

From the Figure 2, it is clear that, the voltage at node '*a*' is zero.

That is, $v_a = 0$.

Apply Kirchhoff's current law node *b* in Figure 2.

$$\frac{v_b - 25}{15\times10^3} + \frac{v_b - (-25)}{10\times10^3} = 0$$
$$\frac{2v_b - 50 + 3v_b + 75}{30\times10^3} = 0$$
$$5v_b + 25 = 0$$
$$v_b = -5\ \text{V}$$

Therefore, the voltage at node '*b*', v_b is $\boxed{-5\ \text{V}}$.

Step 6 of 6

Calculate the voltage between node a and b.

$$\begin{aligned} v_{ab} &= v_a - v_b \\ &= 0-(-5) \\ &= 5\ \text{V} \end{aligned}$$

Therefore, the voltage between node a and b, v_{ab} is $\boxed{5\ \text{V}}$.